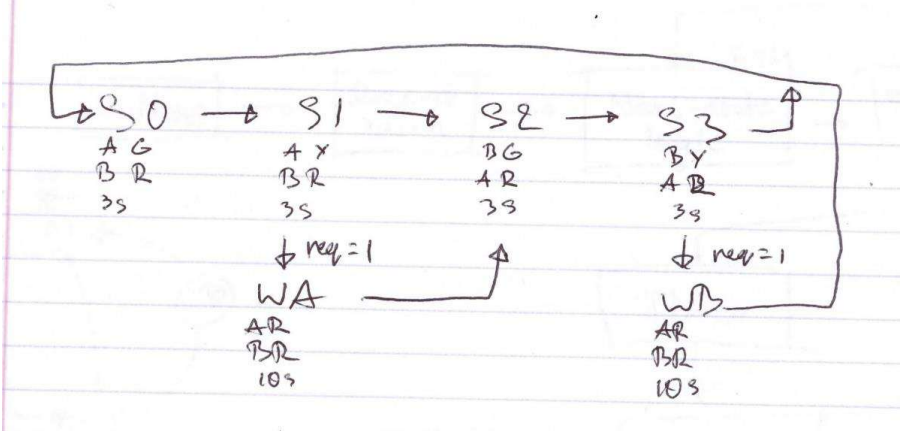
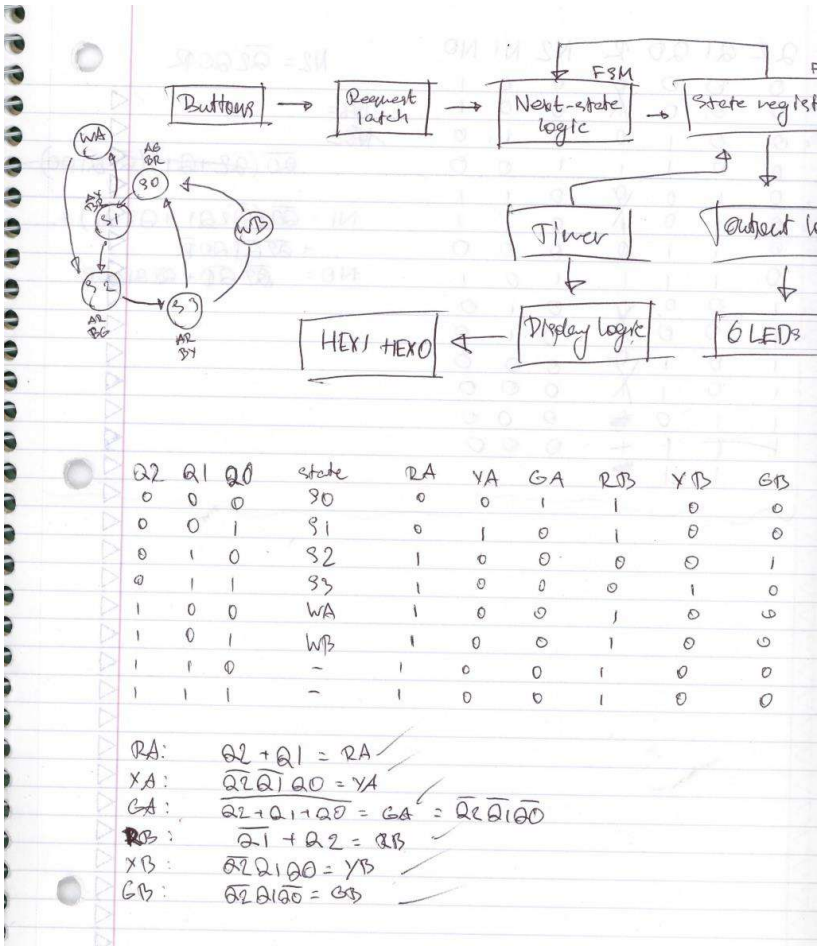
Khan 2101 Final Project Progress Images

Rough Notes

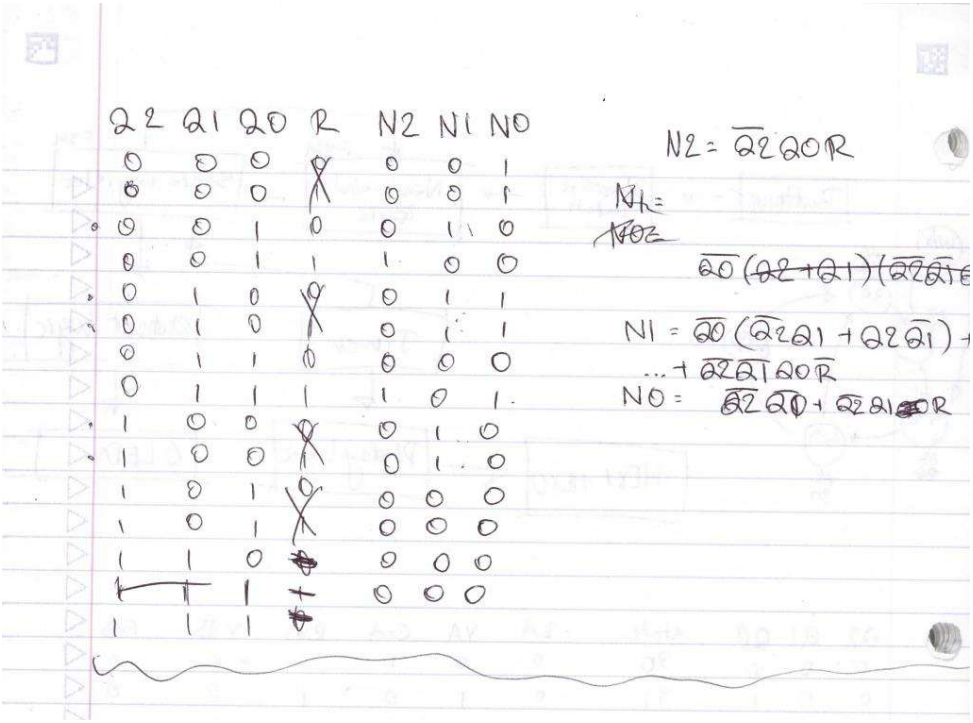


State diagram in notebook



Q2	Q1	Q0	state	RA	YA	GA	RB	YB	GB
0	0	0	S0	0	0	1	1	0	0
0	0	1	S1	0	1	0	1	0	0
0	1	0	S2	1	0	0	0	0	1
0	1	1	S3	1	0	0	0	1	0
1	0	0	WA	1	0	0	1	0	0
1	0	1	WB	1	0	0	1	0	0
1	1	0	–	1	0	0	1	0	0
1	1	1	–	1	0	0	1	0	0

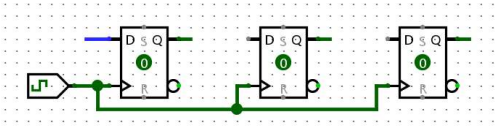
Circuit flowchart, LED output logic table & equations



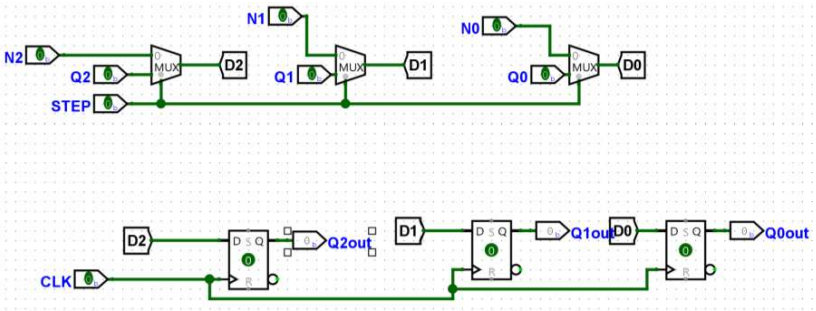
Next-state table & equations

Circuit Progress

Additional notes can be found in "README.md"



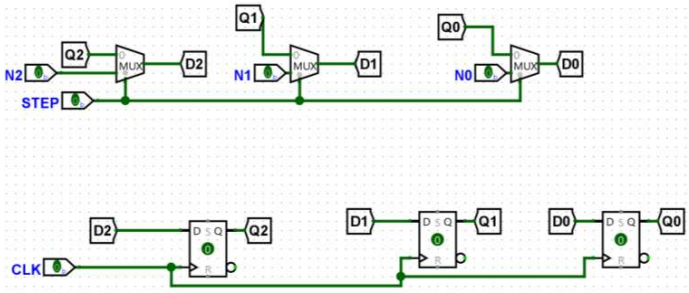
State register flip flops



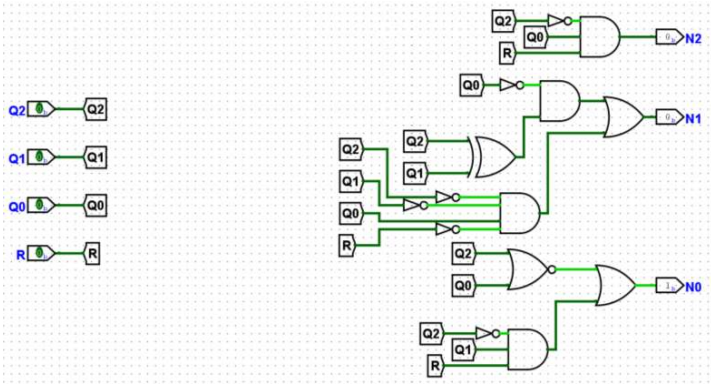
State register local I/O

I tried using inputs and outputs for both the MUX input and the flip-flop output to test the subcircuits functionality, but this was more difficult than testing it on the main

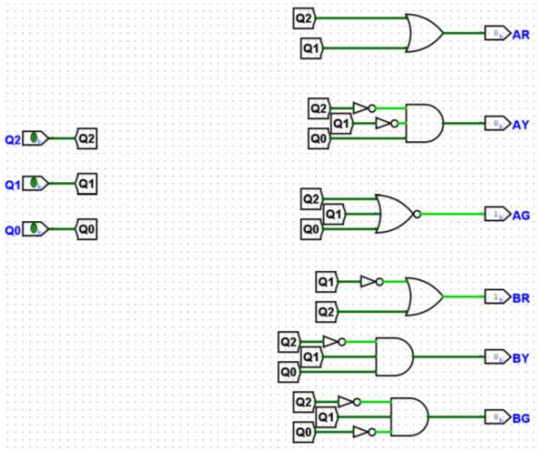
circuit with tunnels within the subcircuit. I had also connected my next-state values to the 0 input of my mux, and the current state values to the 1 inputs.



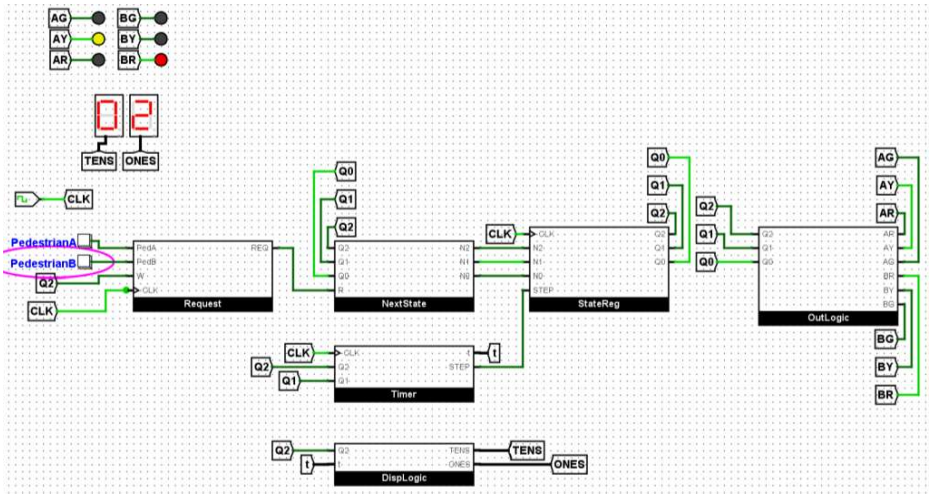
State register tunnels and external I/O connections



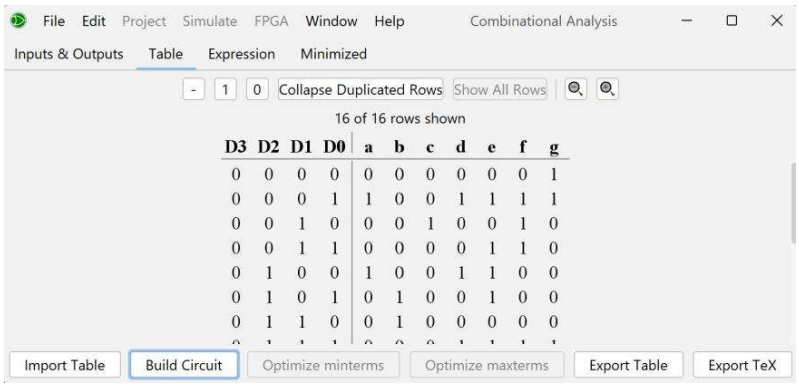
Next-state logic. This and the output logic was fairly simple and worked first try.



Output logic

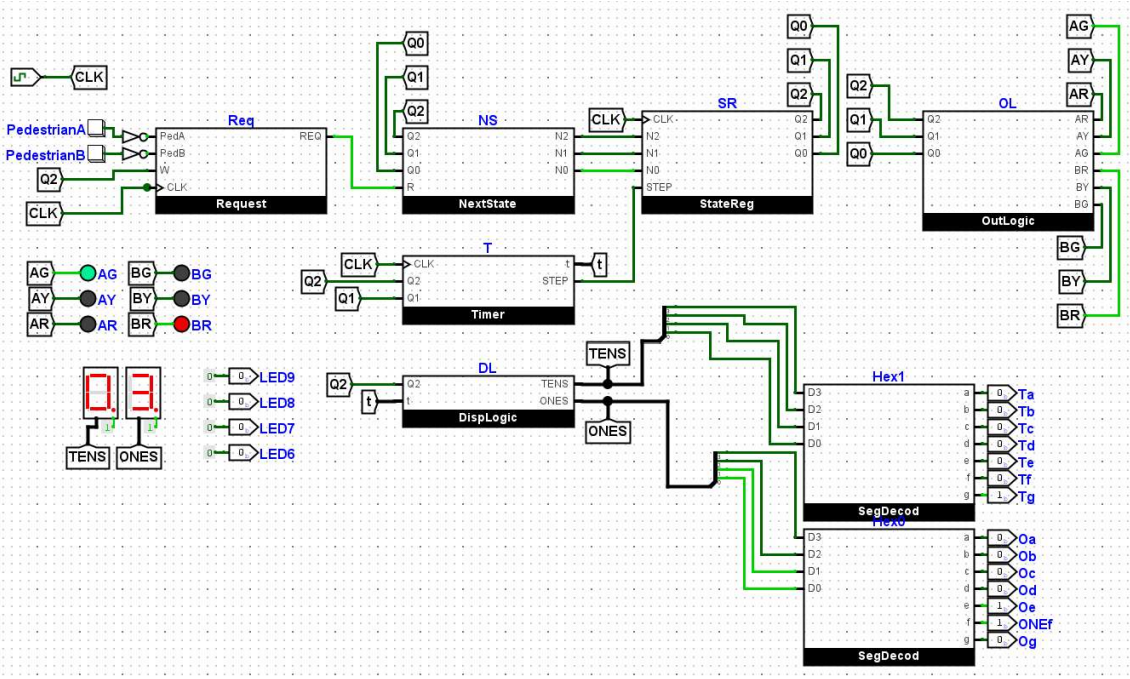


Main circuit first test



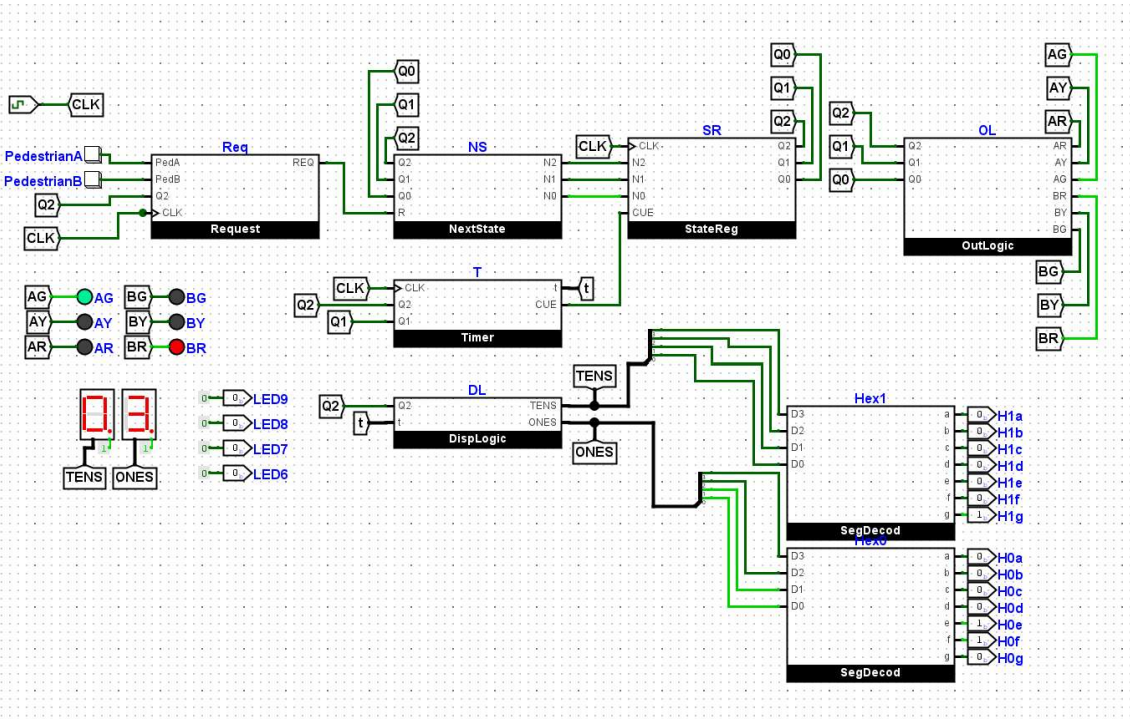
D3	D2	D1	D0	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0	0	1
0	0	0	1	1	0	0	1	1	1	1
0	0	1	0	0	0	1	0	0	1	0
0	0	1	1	0	0	0	0	1	1	0
0	1	0	0	1	0	0	1	1	0	0
0	1	0	1	0	1	0	0	1	0	0
0	1	1	0	0	1	0	0	0	0	0

Combinational analysis tool to build 7-segment decoder

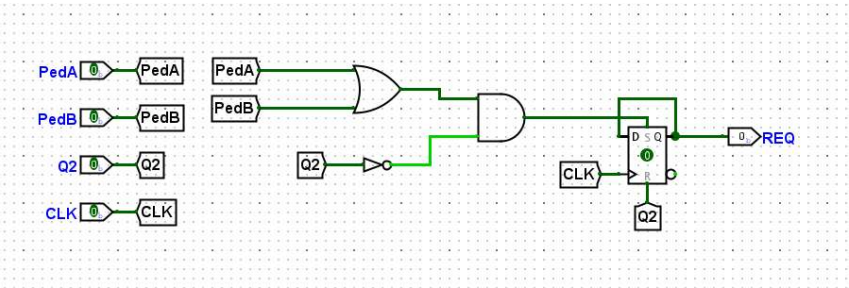


Main with NOT gates. The labels of the decoder output were written quickly to test and did not look very nice.

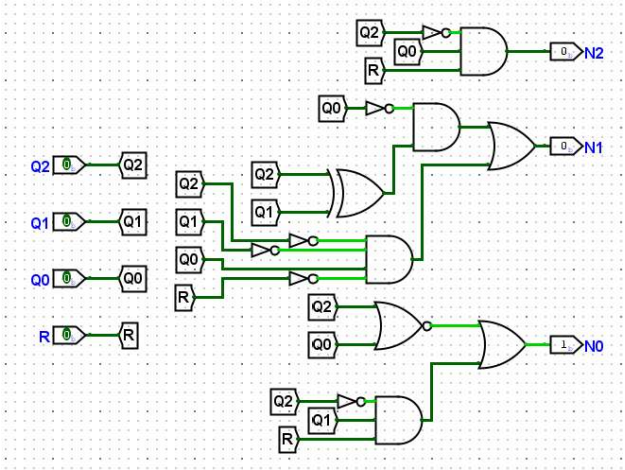
Final circuit



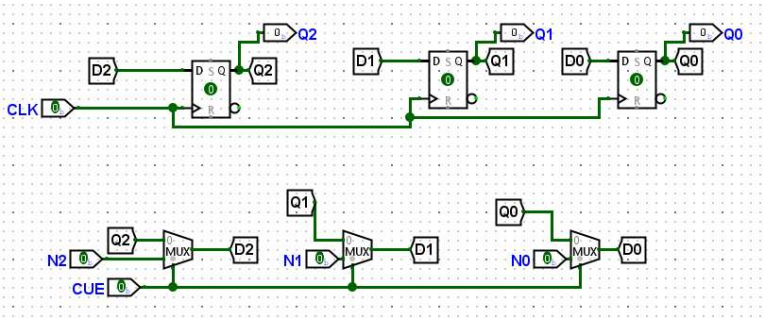
Main circuit with finalised segment titles and passive-on-high buttons.



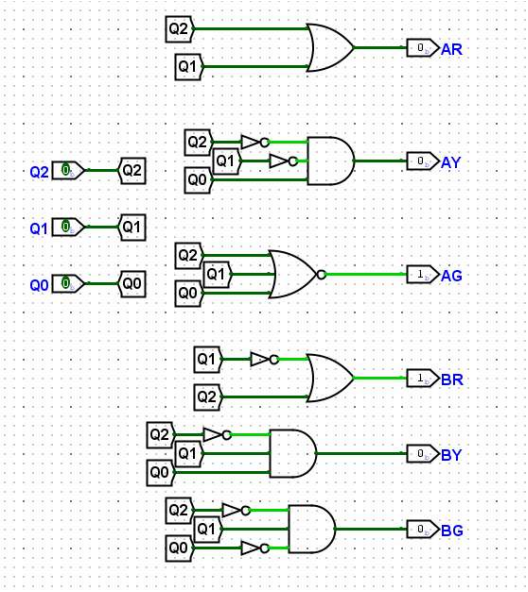
Request subcircuit



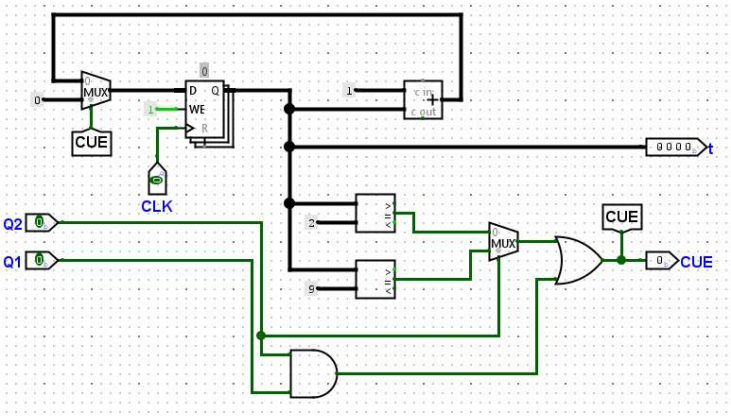
Next-state subcircuit



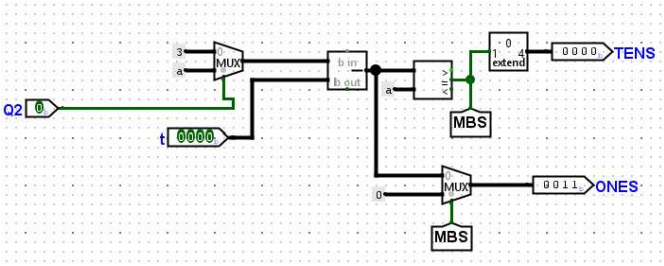
State-register subcircuit



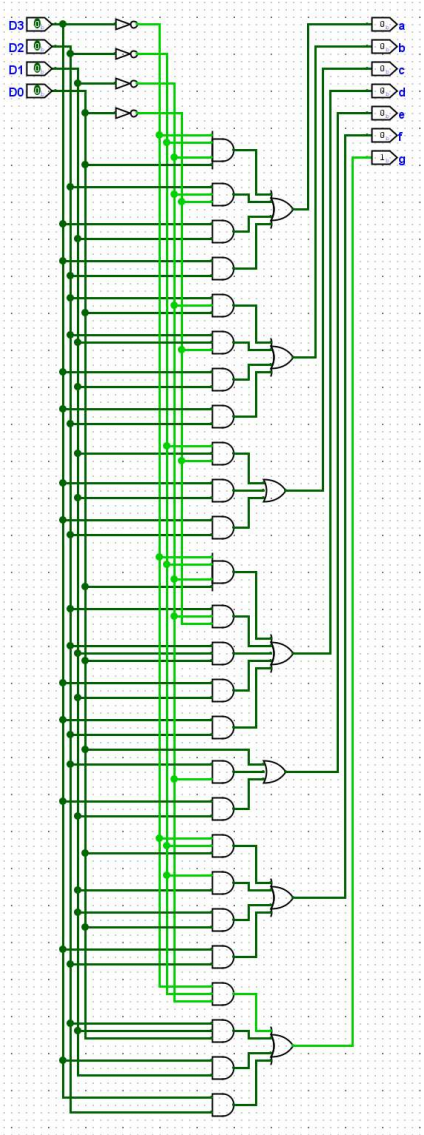
Light output subcircuit



Timer subcircuit



Hex display subcircuit



7-segment decoder